


General Framework of Bayesian Regression

Consider a training dataset

$$\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N,$$

consisting of N observations, where $\mathbf{x}_i \in \mathbb{R}^p$ denotes the p -dimensional predictor vector associated with the i -th observation, and $y_i \in \mathbb{R}$ denotes the corresponding scalar response. Define the design matrix $\mathbf{X}$, response vector $\mathbf{y}$, and regression coefficient vector $\mathbf{w}$ as

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_N^\top \end{bmatrix} \in \mathbb{R}^{N \times p}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix} \in \mathbb{R}^N, \quad \mathbf{w} = \begin{bmatrix} w_1 \\ \vdots \\ w_p \end{bmatrix} \in \mathbb{R}^p.$$

Given the predictor vector $\mathbf{x}_i$ and the regression coefficient vector $\mathbf{w}$, the response y_i is characterized by the conditional distribution

$$p(y_i \mid \mathbf{x}_i, \mathbf{w}).$$

Assuming that the observations are conditionally independent given $\mathbf{w}$, the likelihood of the observed response vector is

$$p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}) = \prod_{i=1}^N p(y_i \mid \mathbf{x}_i, \mathbf{w}).$$

In the Bayesian framework, the regression coefficient vector $\mathbf{w}$ is treated as a random variable and assigned a prior distribution $p(\mathbf{w})$. Given the observed dataset $\mathcal{D} = (\mathbf{X}, \mathbf{y})$, Bayes' theorem gives

$$p(\mathbf{w} \mid \mathcal{D}) = p(\mathbf{w} \mid \mathbf{X}, \mathbf{y}) = \frac{p(\mathbf{w}, \mathbf{X}, \mathbf{y})}{p(\mathbf{X}, \mathbf{y})}.$$

Using

$$p(\mathbf{X}, \mathbf{y}) = p(\mathbf{y} \mid \mathbf{X})p(\mathbf{X}),$$

the posterior distribution can be written as

$$p(\mathbf{w} \mid \mathbf{X}, \mathbf{y}) = \frac{p(\mathbf{w}, \mathbf{y} \mid \mathbf{X})}{p(\mathbf{y} \mid \mathbf{X})} = \frac{p(\mathbf{y} \mid \mathbf{X}, \mathbf{w})p(\mathbf{w} \mid \mathbf{X})}{p(\mathbf{y} \mid \mathbf{X})}.$$

Assuming that the prior distribution of $\mathbf{w}$ is independent of the design matrix $\mathbf{X}$, i.e.,

$$p(\mathbf{w} \mid \mathbf{X}) = p(\mathbf{w}),$$



the posterior distribution becomes

$$\boxed{p(\mathbf{w} \mid \mathbf{X}, \mathbf{y}) = \frac{p(\mathbf{y} \mid \mathbf{X}, \mathbf{w})p(\mathbf{w})}{p(\mathbf{y} \mid \mathbf{X})}}. \quad (1)$$

The denominator

$$\boxed{p(\mathbf{y} \mid \mathbf{X}) = \int p(\mathbf{y} \mid \mathbf{X}, \mathbf{w})p(\mathbf{w})d\mathbf{w}} \quad (2)$$

is referred to as the marginal likelihood, or model evidence.

Since it does not depend on $\mathbf{w}$, the posterior distribution can be expressed up to a proportionality constant as

$$\boxed{p(\mathbf{w} \mid \mathbf{X}, \mathbf{y}) \propto p(\mathbf{y} \mid \mathbf{X}, \mathbf{w})p(\mathbf{w})}, \quad (3)$$

where $p(\mathbf{y} \mid \mathbf{X}, \mathbf{w})$ represents the likelihood and $p(\mathbf{w})$ represents the prior distribution. Therefore, the fundamental principle of Bayesian inference can be summarized as

$$\boxed{\text{Posterior} \propto \text{Likelihood} \times \text{Prior}}.$$

The objective of Bayesian regression is to infer the posterior distribution of the regression coefficient vector $\mathbf{w}$ given the observed dataset $\mathcal{D}$, and subsequently to obtain the posterior predictive distribution of the response y^* for a new observation with predictor vector $\mathbf{x}^*$.

$$p(\mathbf{y}^* | \mathbf{X}^*, \mathcal{D}) = \int p(\mathbf{y}^* | \mathbf{X}^*, \mathbf{w})p(\mathbf{w} | \mathcal{D})d\mathbf{w}.$$